

EDMS 657 HW 1

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1.

The dean of Southeastern State University is interested in the relation between students' grades and their part-time work. Data from 14 students who hold part-time jobs were collected on number of hours worked per week (X) and last semester's grade-point average (Y).

a) [3 points]

Using the method of choice (e.g., hand work, spreadsheet, R...), compute the variances of X and Y. Also compute their covariance and their correlation. Show work or paste results. Then write out your answers all together.

```
ssu_data <- data.frame(x=c(18,7,27,10,13,17,19,22,18,22,15,26,19,32),
                       y=c(3.2,3.2,2.4,3.5,3.1,2.9,2.7,2.1,3,2.5,3.3,2.3,2.3,1.8))
c(Var_X = var(ssu_data$x),
  Var_Y = var(ssu_data$y),
  Cov=cov(ssu_data$x,ssu_data$y),
  Cor=cor(ssu_data$x,ssu_data$y))
```

```
##      Var_X      Var_Y      Cov      Cor
## 44.8406593  0.2609341 -2.9587912 -0.8649937
```

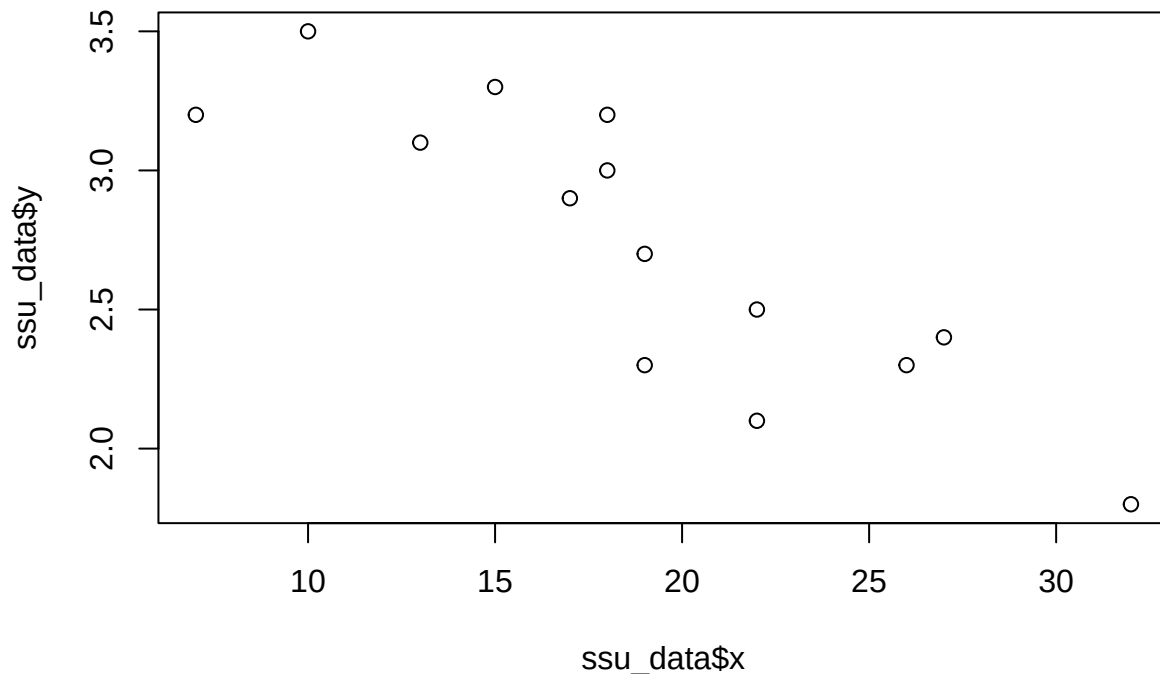
b) [1 point]

If one were able to plot the standardized variable vectors Z_x and Z_y in person space, what would be the angle between those two vectors? Show your brief work.

```
acos(cor(ssu_data$x,ssu_data$y)) * 180/pi
```

```
## [1] 149.882
```

```
plot(ssu_data$x,ssu_data$y)
```



Using the formula $r_{xy} = \cos(\theta_{z_x z_y})$ and inverting to get $\theta = \cos^{-1}(r)$, we can simply plug the correlation from part 1 into the R function for inverse cosine (then multiplying by $180/\pi$ to convert to degrees), obtaining the answer displayed in the output. The value makes sense because it is well above 90 degrees, representing a strong negative correlation which can also be seen in the scatter plot of the data.

2. [2 points]

If two standardized variable vectors have an angle of 85 degrees between them when plotted in person space, what is the correlation between the two variables? Why does this make sense, logically?

$\cos(85^\circ) = 0.0871$, which is a very small positive correlation (in practice, the two variables would likely be deemed completely unrelated). This makes intuitive sense because the components of the vectors are essentially taking the vectors in unrelated directions. Another way to look at it is that knowing the location of one vector gives little to no information about the other (which is also the intuition behind correlation in general). Highly correlated variables would have a smaller angle between them, meaning they are closer together, which also means that knowing the location of one allows a more “confident” estimation of where the other is. On the other end, if the vectors are opposite, we can still be “confident” in the position of one given the other by simply “reversing” the one we know. However, an angle near 90 degrees gives little to no indication of position, as the unknown position could be virtually anywhere around the imaginary circle. More closely-related variables allow the possible position of the unknown vector to be narrowed down quite significantly. That is, we could say “it must be around this *neighborhood*, considering its correlation with the known vector.”

3. [2 points]

Taking a geometric approach (e.g., shish kebob skewers in styrofoam balls, pencils in an apple, etc.), which of $\{-0.2, 0.0, 0.2, 0.8\}$ is NOT possible for r_{32} given that $r_{21} = -0.9$ and $r_{31} = 0.1$? Explain.

Since variables 1 and 2 are strongly negatively correlated ($r_{21} = -0.9$), but variables 1 and 3 are fairly uncorrelated ($r_{31} = 0.1$), it follows that the third and second variables cannot be very highly correlated. Therefore, we can logically rule out the value of 0.8 for r_{32} . Thinking of the correlations geometrically, we are given that variables 1 and 2 have standardized vectors pointing in opposite directions (forming a large angle, near 180 degrees), while variables 1 and 3 have standardized vectors that are near-orthogonal (an angle around 90 degrees). It is impossible for variables 3 and 2 to be highly correlated (or in the same “neighborhood”, as described before), and 0.8 is the only option that represents a small angle between the associated variable vectors. Large angles, though, as described by the other values, are possible.

```
cbind(r21=acos(-0.9) * 180/pi,
      r31=acos(0.1) * 180/pi)
```

```
##           r21      r31
## [1,] 154.1581 84.26083
```

```
sapply(c(-0.2,0,0.2,0.8),acos) * 180/pi
```

```
## [1] 101.53696 90.00000 78.46304 36.86990
```

Additionally, by the “Each pair of angles should sum to a value no less than the third” rule, we can rule out 0.8 mathematically because upon converting the correlations to radians and then degrees, we see $r_{21} : \cos^{-1}(-0.9) \times 180/\pi = 154.15$, $r_{31} : \cos^{-1}(0.1) \times 180/\pi = 84.26$. If $r_{32} : \cos^{-1}(0.8) \times 180/\pi = 36.87$, the sum of the angles for r_{31} and r_{32} is less than the angle for r_{21} ($84.26 + 36.87 = 121.13 < 154.15$).

4.

Imagine that the SATV (verbal) test scores have a published reliability coefficient of .80, and that the SATM (math) scores have a published reliability coefficient of .90. The observed correlation between the scores is .50.

a) [1 point] What would we expect the correlation to be between SATV scores and true verbal ability?

```
sqrt(.8)
```

```
## [1] 0.8944272
```

Using $\rho_{T_v, v} = \sqrt{\text{rel}(v)}$, where T_v represents true verbal ability, v on its own represents SATV scores, and $\text{rel}()$ is an abbreviation for reliability.

b) [2 points]

What correlation would we expect between true verbal ability and true math ability if we could measure true verbal ability and true math ability directly?

```
0.5/(sqrt(0.8)*sqrt(0.9))
```

```
## [1] 0.5892557
```

The above used the formula

$$\rho_{T_v, T_m} = \frac{\rho_{vm}}{\sqrt{\text{rel}(s)}\sqrt{\text{rel}(m)}} = \frac{0.5}{\sqrt{0.8}\sqrt{0.9}} = 0.58925$$

Where T_v and T_m represent true verbal and true math scores, respectively, and ρ_{vm} is the observed correlation between math and verbal scores. $\text{rel}()$ is an abbreviation of reliability.

The wording of the question almost seems that an answer could be to take a sample of “true” scores for each category, then simply compute the correlation between them.

5.

Imagine that the Maryland state assessment of reading ability has four subtests to it, correlating in the matrix as follows for V_1, V_2, V_3, V_4 :

$$\begin{bmatrix} 1 & & & \\ .38 & 1 & & \\ .42 & .53 & 1 & \\ .28 & .47 & .44 & 1 \end{bmatrix}$$

a) [2 points]

Assume that these four tests all attempt to measure (i.e., are influenced by) the same underlying latent skill, and then have unique skills associated with each. Draw a path diagram representing this theory.

b) [4 points]

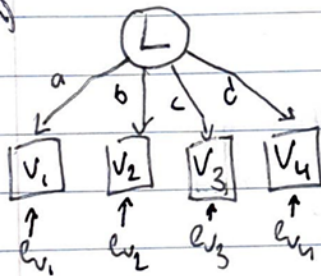
Using Spearman’s method of triads, estimate the standardized effect of true reading ability on V1, on V2, on V3, and on V4 (i.e., the correlation between true reading ability and test scores on each of V1, V2, V3, and V4).

c) [3 points]

Based on your triad solution, determine the reproduced correlation matrix and the residual correlation matrix.

HW1 #5

a)



• L for "latent"

• a, b, c, d: Strength or relationship between L and

V_1, V_2, V_3, V_4 scores, each respectively

• e_i : unique skill associated with each of the $i=1, \dots, 4$ tests

b)

$$\begin{bmatrix} 1 & & & \\ .38 & 1 & & \\ .42 & .93 & 1 & \\ .28 & .47 & .44 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} L^2 & & & \\ ab & b^2 & & \\ ac & bc & c^2 & \\ ad & bd & cd & d^2 \end{bmatrix}$$

implies

$$\frac{4-1}{2} = \frac{3}{2} = 1.5$$

$$\frac{3-2}{2} = 0.5$$

$$a^2 = \frac{ab \cdot ac}{bc} \quad \text{or} \quad \frac{ab \cdot ad}{bd} \quad \text{or} \quad \frac{ac \cdot ad}{cd}$$

$$= \frac{.38 \cdot .42}{.93} \quad \text{or} \quad \frac{.38 \cdot .28}{.47} \quad \text{or} \quad \frac{.42 \cdot .28}{.44}$$

$$= 0.3011 \quad \text{or} \quad 0.2264 \quad \text{or} \quad 0.2678$$

Stable way:

$$a^2 = \frac{.38 \cdot .42 + .38 \cdot .28 + .42 \cdot .28}{.93 + .47 + .44}$$

$$= \frac{0.3936}{1.44} = 0.2664$$

$$\Rightarrow a = .5161$$

$$b: b^2 = \frac{a_5 \cdot b_6}{a_6} \text{ or } \frac{b_6 \cdot b_2}{b_2} \text{ or } \frac{a_5 \cdot b_2}{a_2}$$

$$\rightarrow \frac{0.38 \cdot .53 + .53 \cdot .47 + .38 \cdot .47}{.42 + .44 + .28} = \frac{.6291}{1.14} = .5518$$

$$\Rightarrow b = .7428$$

$$c: c^2 = \frac{a_6 \cdot b_6}{a_5} \text{ or } \frac{a_6 \cdot c_2}{a_2} \text{ or } \frac{b_6 \cdot c_2}{b_2}$$

$$\rightarrow \frac{.42 \cdot .53 + .38 \cdot .44 + .53 \cdot .44}{.38 + .28 + .47} = \frac{.623}{1.13} = .5513$$

$$\Rightarrow c = .7425$$

$$d: d^2 = \frac{a_6 \cdot c_2}{a_6} \text{ or } \frac{a_6 \cdot b_2}{a_5} \text{ or } \frac{b_2 \cdot c_2}{b_6}$$

$$\rightarrow \frac{.28 \cdot .44 + .28 \cdot .47 + .47 \cdot .44}{.42 + .38 + .53} = \frac{.4616}{1.33} = .3471$$

$$\Rightarrow d = .5891$$

$$a = .5161, b = .7428, c = .7425, d = .5891$$

$$C) \begin{bmatrix} a^2 & & & \\ a_5 & b^2 & & \\ a_6 & b_6 & c^2 & \\ a_2 & b_2 & c_2 & d^2 \end{bmatrix} = \begin{bmatrix} .2664 & & & \\ .3833 & .5518 & & \\ .3832 & .5518 & .5513 & \\ .304 & .4376 & .4371 & .3471 \end{bmatrix}$$

reproduced matrix

observed - reproduced:

$$\begin{bmatrix} 0.0033 & & & \\ -0.0368 & 0.0245 & & \\ 0.024 & -0.0324 & -0.0021 & \end{bmatrix}$$

residual correlation matrix
pretty small?

Why is the diagonal so far off though?